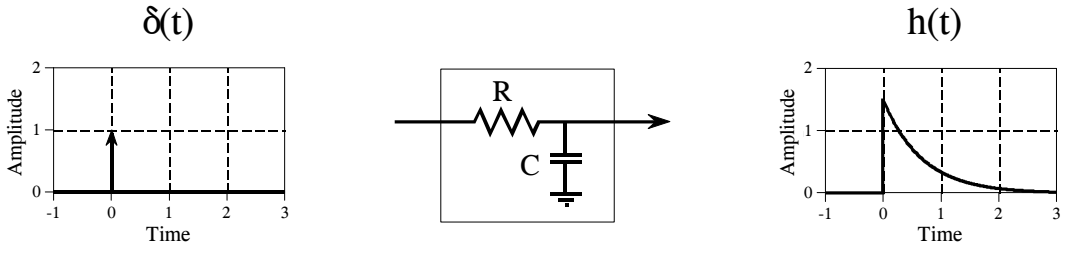

FIGURE 13-4
Example of a continuous linear system. This electronic circuit is a low-pass filter composed of a single resistor and capacitor. The impulse response of this system is a one-sided exponential.

impulse response of this system is broken into two sections, each represented by an equation:

$$h(t) = 0 \qquad \text{for } t < 0$$

$$h(t) = \alpha e^{-\alpha t} \qquad \text{for } t \geq 0$$

where $\alpha = 1/RC$ (R is in ohms, C is in farads, and t is in seconds). Just as in the discrete case, the continuous impulse response contains complete information about the system, that is, how it will react to all possible signals. To pursue this example further, Fig. 13-5 shows a square pulse entering the system, mathematically expressed by:

$$x(t) = 1 \qquad \text{for } 0 \leq t \leq 1$$

$$x(t) = 0 \qquad \text{otherwise}$$

Since both the input signal and the impulse response are completely known as mathematical expressions, the output signal, $y(t)$, can be calculated by evaluating the convolution integral of Eq. 13-1. This is complicated by the fact that both signals are defined by *regions* rather than a single

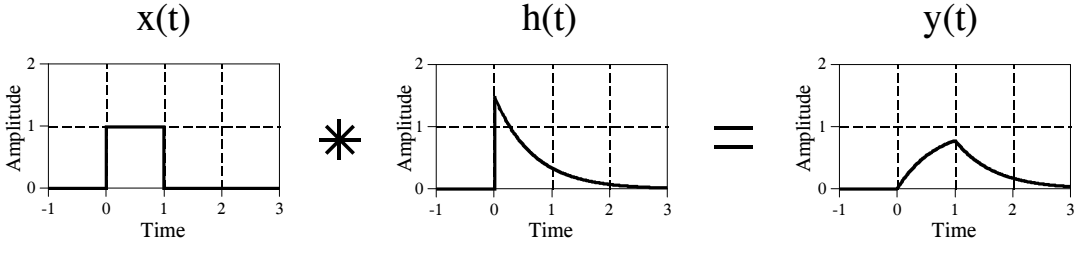

FIGURE 13-5
Example of continuous convolution. This figure illustrates a square pulse entering an RC low-pass filter (Fig. 13-4). The square pulse is convolved with the system's impulse response to produce the output.